Section A - Strings

Q1. Reverse a string without using slicing.

```
text = "python"

reversed_text = ""

for character in text:

    reversed_text = character + reversed_text

print(reversed_text)
```

Q2. Find the first non-repeating character in a string.

```
text = "programming"
for character in text:
    count = 0
    for item in text:
        if character == item:
            count += 1

    if count == 1:
        print(character)
        break
```

Q3. Check if a string is a palindrome.

```
text = "madam"
reversed_text = ""

for character in text:
    reversed_text = character + reversed_text

if text == reversed_text:
    print(True)
else:
    print(False)
```

Q4. Count the frequency of each character in a string.

```
text = "hello"
frequency = {}
for character in text:
    if character in frequency:
        frequency[character] += 1
    else:
        frequency[character] = 1
```

```
for character, count in frequency.items():
    print(character, ":", count)
```

Q5. Remove duplicate characters from a string while preserving order.

```
text = "programming"
result = ""

for character in text:
    if character not in result:
        result += character

print(result)
```

Section B - Lists

Q6. Remove duplicates from a list without using set().

```
numbers = [1, 2, 2, 3, 4, 4]
unique_numbers = []

for number in numbers:
    if number not in unique_numbers:
        unique_numbers.append(number)

print(unique_numbers)
```

Q7. Find the second largest number in a list.

```
numbers = [10, 20, 5, 30, 25]
unique_numbers = []

for number in numbers:
    if number not in unique_numbers:
        unique_numbers.append(number)

unique_numbers.sort(reverse=True)
print(unique_numbers[1])
```

Q8. Find all duplicate elements in a list.

```
numbers = [1, 2, 3, 2, 4, 5, 1]
duplicates = []

for number in numbers:
    if numbers.count(number) > 1 and number not in duplicates:
        duplicates.append(number)

print(duplicates)
```

Q9. Rotate a list by K positions

```
numbers = [1, 2, 3, 4, 5]
k = 2

k = k % len(numbers)
rotated_list = numbers[-k:] + numbers[:-k]

print(rotated_list)
```

Q10. Find the intersection of two lists.

```
list1 = [1, 2, 3, 4]
list2 = [3, 4, 5, 6]
intersection = []

for number in list1:
    if number in list2 and number not in intersection:
        intersection.append(number)

print(intersection)
```

Section C - Dictionary

Q11. Count frequency of elements in a list using a dictionary.

```
numbers = [1, 2, 2, 3, 3, 3]
frequency = {}

for number in numbers:
    if number in frequency:
        frequency[number] += 1
    else:
        frequency[number] = 1

print(frequency)
```

Q12. Find the key having the maximum value.

```
data = {"A": 100, "B": 500, "C": 300}
max_key = None
max_value = None

for key, value in data.items():
    if max_value is None or value > max_value:
        max_value = value
        max_key = key

print(max_key)
```

Q13. Reverse a dictionary.

```
data = {"a": 1, "b": 2}
reversed_dict = {}

for key, value in data.items():
    reversed_dict[value] = key

print(reversed_dict)
```

Q14. Merge two dictionaries.

```
d1 = {"a": 1}
d2 = {"b": 2}

merged = d1.copy()
merged.update(d2)

print(merged)
```

Q15. Count word frequency in a sentence using a dictionary.

```
sentence = "python is good python is easy"
words = sentence.split()
frequency = {}

for word in words:
    if word in frequency:
        frequency[word] += 1
    else:
        frequency[word] = 1

print(frequency)
```

Section D - Loops

Q16. Print the given pattern.

```
for i in range(2):
    print("*", end="")
```

Q17. Print multiplication table of a given number.

```
number = 5

for i in range(1, 11):
    print(number, "x", i, "=", number * i)
```

Q18. Find factorial using a for loop.

```
number = 5
factorial = 1

for i in range(1, number + 1):
    factorial = factorial * i

print(factorial)
```

Q19. Find all prime numbers between 1 and 100.

```
for number in range(2, 101):
    is_prime = True

    for divisor in range(2, number):
        if number % divisor == 0:
            is_prime = False
            break

    if is_prime:
        print(number, end=" ")
```

Q20. Generate Fibonacci series up to N terms.

```
n = 8
first = 0
second = 1

for i in range(n):
    print(first, end=" ")
    next_number = first + second
    first = second
    second = next_number
```

Section E - Interview Coding Questions

Q21. Find the first non-repeating number in a list.

```
numbers = [1, 2, 3, 4, 5, 1, 2, 3]

for number in numbers:
    if numbers.count(number) == 1:
        print(number)
        break
```

Q22. Find the Nth non-repeating number in a list.

```
numbers = [1, 2, 3, 4, 5, 1, 2, 3]
n = 2
```

```
non_repeating_count = 0

for number in numbers:
    if numbers.count(number) == 1:
        non_repeating_count += 1

        if non_repeating_count == n:
            print(number)
            break
```

Q23. Check whether two strings are anagrams.

```
word1 = "listen"
word2 = "silent"

if sorted(word1.lower()) == sorted(word2.lower()):
    print(True)
else:
    print(False)
```

Q24. Find the missing number from an array.

```
numbers = [1, 2, 3, 5]

for number in range(1, max(numbers) + 1):
    if number not in numbers:
        print(number)
        break
```

Q25. Find the top occurring element in a list.

```
numbers = [1, 2, 2, 3, 3, 3, 4]

top_element = max(numbers, key=numbers.count)
print(top_element)
```